CELULAR AUTONOMA v2

Self-replicating smart-contract cells that pay for themselves

Findings, proposal, exploratory tests & results
from orbitmechanic/cellular_autonoma
Analysis + v2 reference implementation - 2026

39 passing tests - 6 issues (#12-#17) - PR #18 (+2024 / -263)

The brief

- Build a life-like 'cell' in Solidity: an address-linked collection of organelles on an EVM chain.
- Nucleus holds identity + linked modules; Chloroplast stores energy (= capital); and so on.
- Goal: a contract colony that self-replicates with NO human intervention.
- It must find its own 'gas' for memory + processing once launched.
- It must be capital-harvestable by the colony root deployer - 'or why bother?'.
- v1 stalled on gas costs and gaps that needed better code. Can we do better?

The hard constraint (why v1 stalled)

On the EVM, a contract can NEVER send its own transaction. Only an external account (or smart-account relayer) pays gas and initiates.

Three ways to relax it

- 1. Cheap chain: L2 / near-free so a division costs cents, not dollars.
- 2. Keeper economics: the cell pays whoever triggers it, from capital it earned.
- 3. ERC-4337 paymasters: a bundler pays gas, cell settles in tokens.

v1 did none of these fully.

L1 division cost (measured):
v1 ~$82 v2 ~$49
So the loop could never pay its own gas and the colony starved in the first few generations.

Findings: six verified v1 failures

Each is reproduced by an executable characterization test in the PR.

##12	Dead capital	Keeper pays 100% gas; estimate is hoarded, never reimbursed
##14	Proxy-depth bloat	Clones live instances -> +1 DELEGATECALL each generation
##14	Identity bloat	'...copy copy' string grows unbounded (10->35 bytes/gen)
##15	No harvest	Capital never reaches the root owner (the whole point)
##16	Broken gate	Membership check reverts wrong; any organelle can withdraw; daughters inherit the mother's metabolism
##17	Tests broken	ethers v5 code on a v6 toolchain: 0 passing; references a non-existent factory

Exploratory result: inherited gas bloat

v1 identity() call gas by generation

```
gen1:  23997 gas
gen2:  26675 gas
gen3:  29467 gas
gen4:  32281 gas
gen5:  34980 gas
gen6:  37971 gas
+58%  by gen 6, still climbing
```

v2 getBalance() call gas by generation

```
gen1:  23882 gas
gen2:  23882 gas
gen3:  23882 gas
gen4:  23882 gas
gen5:  23882 gas
gen6:  23882 gas

FLAT - clone from template, not instance
```

v2 kills the compounding by cloning immutable TEMPLATES in Colony.birth(). A colony no longer gets pricier to run as it ages.

Proposal: chain choice decides viability

One full v2 division ~= 1.08M gas. Projected USD cost per self-replication:

Chain	v1 / div	v2 / div	Self-sustaining?
Ethereum L1	$82.0329	$48.6094	NO - keeper can't profit at scale
Arbitrum One	$0.5469	$0.3241	MARGINAL - viable for high-value cells
OP Mainnet (base)	$0.2734	$0.1620	MARGINAL - viable for high-value cells
Base	$0.0547	$0.0324	YES - cheap + huge DeFi/AI ecosystem
Polygon PoS	$0.0273	$0.0162	YES - cheap + huge DeFi/AI ecosystem
Sonic	$0.0006	$0.0004	NEAR-FREE - best raw economics

Recommendation: launch on Base (ecosystem to earn from) and/or Sonic (near-free division). Same EVM bytecode both.

The v2 organism: Colony + organelles

Colony	The one trusted mitosis engine. Clones organelles from immutable templates and wires each daughter. plantRoot / birth / harvest / graft / apoptosis.
Nucleus	Identity (serial + generation), capability-role registry, lineage tracked both directions (parent + children).
EnergyStore	The 'food'. Per-payer ledger, role-gated withdrawals, pull-payment keeper credits, remitUp tithes.
Chloroplast	Metabolism + replication. Gated replicate(): cooldown, generation cap, reserve floor. Pays the keeper.

Economic autonomy (the achievable version)

A cell still can't send its own tx - but ANYONE can trigger it and get paid from the cell's own earned food. Triggering becomes a job.

Repayment loop

- Keeper calls replicate() from an EOA / Automation / bot.
- Cell credits the keeper's gas (budgeted, capped by maxRefundCap/Bps).
- Pull-payment ledger: keeper claims later; harvest can't strand it.
- On Base/Sonic the credit is tiny, so division self-funds.

Harvest to the root (the 'or why bother')

- birthFee per division accrues to the Colony treasury.
- birth tithe: a slice of each daughter flows back up the tree.
- harvestColony: gas-bounded BFS sweeps every cell to the owner.
- apoptosis: retire a cell, reclaim its capital.

What cells do - and how they earn

A cell must earn more food than it burns paying its metabolism, or it dies. v2 ships the per-payer ledger + RESERVE/graft slots these plug into.

Metabolism (income)

- Keeper-for-hire relayer (Chainlink Automation / Gelato).
- Yield: idle food into Aave/Morpho/LSTs = true photosynthesis.
- Service endpoints sold to AI agents (attestation, escrow, clocks).
- Storage / registry rent for other contracts.

Multicellular organisms

- Divide labour: siblings carry different RESERVE/service organelles.
- Shared genome: template clones -> 1000-cell colony, linear cost.
- Apoptosis retires sick cells, keeps the organism healthy.
- Colony treasury + membranes are counterparties to protocols, bots and humans.

Tests & results

39	6	PR#18	41%
tests passing	issues filed #12-#17	+2024 / -263	cheaper division

Suites

- 3 x v1 characterization suites - prove each flaw is real (and quantify the bloat).
- v2-cell.test.js - genesis, roles, mitosis, template-clone flatness, generation cap, symbiosis, apoptosis.
- v2-economics.test.js - keeper whole, refund caps, tithes, harvestColony tree-walk, fee treasury.
- scripts/gas-study.js regenerates gas-study.json (the bloat + cross-chain tables in this deck).

v1 vs v2, at a glance

Aspect	v1	v2 (this PR)
Who divides	caller pays gas, unreimbursed	cell reimburses keeper (pull)
Cloning	live instances (bloat)	immutable templates (flat)
Identity	'...copy copy' unbounded	serial + generation, cell#N
Fund authority	any organelle, revert bug	role-gated METABOLISM/Colony
Daughter wiring	shares mother's metabolism	fresh trio, assembled by Colony
Harvest	none	tithe + fee + tree sweep
Division limits	none	cooldown + gen-cap + reserve
Tests	0 passing (broken)	39 passing
Division gas	1.82M	1.08M (-41%)

Recommendation & next steps

Ship the same EVM bytecode to Base and/or Sonic. Merge PR #18, then build the income organelles.

- 1. Deploy Colony to a Base Sepolia testnet; publish keeper-reimbursement params.
- 2. Build Membrane (metered per-payer income) + a yield RESERVE - the real photosynthesis.
- 3. Wrap Chloroplast as an ERC-4337 smart account so a paymaster pays gas in tokens.
- 4. Governance organelle -> owner-less colonies; reputation scoring for which cells to propagate.
- 5. Turntable run: a colony living on its own earned income for N days; measure harvest-to-root vs gas burned.

Honest bottom line: 'self-replicating without humans' is unachievable literally; economically autonomous + harvestable IS achievable, and now runs green.

Artifacts

github.com/orbitmechanic/cellular_autonoma

Issues (found)

- #13 autonomy / gas blocker (design)
- #12 keeper reimbursement
- #14 proxy + identity bloat
- #15 harvest to root
- #16 roles / metabolism / gate
- #17 broken test suite

Shipped (PR #18)

- contracts/v2/ ICell, Nucleus, EnergyStore, Chloroplast, Colony
- 2 v2 suites + 3 v1 characterization suites
- scripts/gas-study.js + gas-study.json
- DESIGN.md (full findings + strategy)
- inline code review (7 notes) on PR #18

Status: PR #18 OPEN for review. Not self-merged (security rewrite).